

NAMED ENTITY RECOGNITION FOR SCANNED IMAGES, AUDIO AND VIDEO

Major Project Phase-III (B)



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Introduction

- ▶ Named Entity Recognition (NER) is an important method in Natural Language Processing (NLP) that helps identify and classify names of people, organizations, locations, dates, and other entities within text.
 - ▶ Originally, NER was used with written text, but now it is being applied to different formats like scanned images, audio, and video.
 - ▶ For scanned images, NER has to handle the tough job of reading handwritten or printed text and linking it to specific categories. This task has become even more crucial with the digitization of historical documents and media.
 - ▶ Applying NER to audio and video involves transcribing spoken words and extracting text from visual content, which presents both challenges and opportunities.
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Problem Statement

- ▶ Named Entity Recognition (NER) is a crucial task in natural language processing that identifies and categorizes entities within text, but it faces challenges with scanned images due to poor image quality, varied fonts, handwriting styles, noise, and distortions, and with audio and video sources due to speech recognition errors and visual analysis difficulties.
 - ▶ This project aims to develop an advanced NER solution that accurately extracts and categorizes named entities from scanned images, audio, and video by using optical character recognition (OCR) for scanned images to convert text into a machine-readable format, speech recognition for audio to transcribe spoken content, and visual analysis for video to identify and extract text fragments, ensuring effective NER across all media formats.
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Literature Survey

Topic Name	Author	Year
OCR and NER for Scanned Documents	A. Deoras and S. Sarkar	2015
Named Entity Recognition in OCR-generated Texts	R.Soualah-Alila and J. Tabbone	2018
Named Entity Recognition for Scanned Images	Anuva Goyal and Kritika	2021
Named Entity Recognition (NER) in NLP Techniques, Tools Accuracy and Performance	Salman Naseer and Sohaib Bin Khalid Alvi	2022
Multimodal Named Entity Recognition for Video and Audio Streams	Bo Li, Zhijian Ou, Dong Yu	2017
Multimodal Named Entity Recognition for Real-world Vide	J. Xu, C. Xiong, C. Liu, D. Song	2019

Proposed System

- ▶ Named Entity Recognition (NER) on scanned images is challenging because it usually involves using Optical Character Recognition (OCR) to extract text from images. Here are some tools and libraries we can use:
 - i) OCR with EasyOCR
 - ii) NER with Spacy, Displacy
 - ▶ For NER on audio and video, specific techniques and tools are required:
 - i) Audio: SpeechRecognition, Wisper
 - ii) Video: Youtube-transcript-api
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Requirements

► Software Requirements :

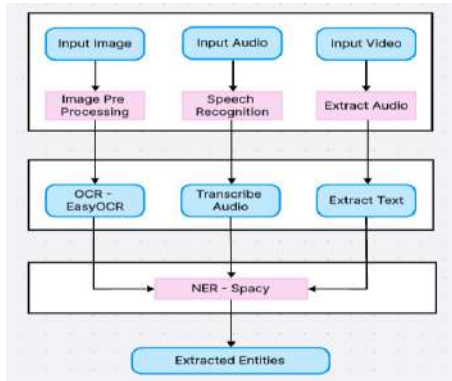
Language : Python

Library : EasyOCR, Spacy, SpeechRecognition, Youtube-transcript-api, hspaceWisper

Tools : Google Colab, Visual Studio

Documentation tool : Overleaf

Implementation



Complete Working of Project

Application

- ▶ Government
 - ▶ Health Care
 - ▶ Financial Services
 - ▶ Banking
 - ▶ Document Analysis and OCR
 - ▶ Voice Assistants and Chatbots
 - ▶ Video Content Analysis
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Work Done for Phase I (A)

Phase I	
Completed Work	Feedback
Text NER	Use Handwritten text image as an input
Extracted Text from images using OCR	To improve Accuracy
Applying NER on Extracted Text	

Work Done for Phase I (B)

Phase I (B)	
Completed Work	Feedback
Taken Handwritten Text Images	To improve Accuracy using customized NER models for specific domains
Extracted text from those image	To convert text images in various languages including Hindi and Marathi, and apply NER
And performed NER	To build Named Entity Recognition in audio and video
Used different pre-trained English language models to improve accuracy	

Work Done for Phase II (A)

Phase II (A)	
Completed Work	Feedback
We have used an audio file as input	Try to solve ambiguity
Then, we extracted text data from the audio file	To improve more accuracy Audio NER
Next, we performed NER on it	And do the tasks which was not completed in previous phase
	To build NER on video

Work Done for Phase II (B)

Phase II (B)	
Completed Work	Feedback
Taken YouTube Video file as input	To improve Accuracy in all Images, Audio and Video
Then extracted its transcript as a text data	
Then used that extracted data for applying NER	

Work Done for Phase III (A)

Phase III (A)	
Completed Work	Feedback
We have created new customized NER entity	Complete the project
Solved ambiguity of domain specific	Start making Report Documentation
We made Technical Paper for our project	

Work Done for Phase III (B)

Phase III (B)	
Completed Work	Feedback
We have completed the implementation of the project.	
We have completed the report as well.	

Interaction with Guide

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Conclusion

- ▶ In conclusion, Named Entity Recognition (NER) on scanned images is a significant advancement in data analysis, enabling us to extract crucial information like names, dates, and places using machine learning techniques.
 - ▶ Initially, we convert images into text using EasyOCR, then utilize tools like spaCy for NER, ensuring accuracy and reliability.
 - ▶ Expanding NER to audio and video with tools such as SpeechRecognition, Youtube-transcript-api, and Wisper enhances our data analysis capabilities across various media formats, facilitating easier information processing and knowledge discovery.
 - ▶ This comprehensive approach allows us to extract entities from images, transcribe spoken words, and analyze video content, ultimately improving our understanding and utilization of data.
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Future Scope

- ▶ In future scope, we are focusing on two main areas to improve Named Entity Recognition. Firstly, we will develop custom NER models tailored for specific domains or languages.
 - ▶ These models will be fine-tuned to work better with domain specific scanned documents, and we will also explore creating new categories for better accuracy.
 - ▶ Secondly, we aim to make NEW work for multiple languages, including Hindi and Marathi, by converting text from image.
 - ▶ Additionally, we are excited to explore NER's potential in extracting entities from handwritten texts. By addressing there areas, we hope to enhance NER's effectiveness across different languages and real-world scenarios.
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Reference

- ▶ <https://www.researchgate.net/publication/357420807-NAMED-ENTITY-RECOGNITION-FOR-SCANNED-IMAGES>
 - ▶ <https://www.researchgate.net/publication/376355415-A-Novel-Technique-for-Handwritten-Text-Recognition-using-Easy-OCR>
 - ▶ <https://www.researchgate.net/publication/372339111-Video-Transcript-Summarizer>
 - ▶ <https://www.researchgate.net/publication/317079194-BEING-A-POLYGLOT-A-CASE-STUDY-OF-A-NINE-LANGUAGES-SPEAKER>
 - ▶ <https://www.researchgate.net/publication/328777063-librosa-Audio-and-Music-Signal-Analysis-in-Python>
 - ▶ <https://dl.acm.org/doi/abs/10.1007/978-981-99-4752-2-50>
 - ▶ <https://arxiv.org/abs/2106.08977/>
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Thank You!
